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Weekly Report

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1. Purpose

This week I reviewed and reproduced the main idea of *Building a Personalized Dialogue System with Prompt-Tuning* (Kasahara et al., 2022). The goal was **not to add a new production model to MMTT**. It was to test a useful boundary: how far can a persona make a dialogue model feel more human, and which parts still require training?

For MMTT, the practical target is **not to maximize an “AI pass rate.”** The target is a **pre-specified witness prompt** for which participants are close to chance when judging AI versus human, while the study still reports both hit rate and false-alarm rate through d' and criterion c. This keeps the research question intact: *where is the signal of humanness written in language, and what does MT preserve or weaken?* A prompt chosen only after repeatedly checking the pass rate would turn the experiment into score chasing and bias the result.

2. Prompt-Tuning Reproduction

2.1 What was reproduced

The reproduction repository(注1) reimplements the paper’s core method with a freely available Japanese persona-dialogue dataset. The base Qwen2.5 model and its token embedding layer are frozen. **Only a 200-token soft persona embedding is trained**; loss is applied only to the answer tokens. Persona sentences initialise the soft prompt, and persona dialogue is mixed 1:1 with short general dialogue. This is a close reproduction of the **method, but not a direct result replication**.

Each of the three selected personas has 946 training pairs (473 persona-related and 473 general) and 52 held-out persona pairs. The shared general evaluation set has 150 pairs. The 0.5B run was trained for 40 epochs; the 3B run for 50 epochs.

表 1　Paper versus local reproduction.

Item	Original paper	Local reproduction
Base model	GPT-2 / GPT-J / HyperCLOVA	Qwen2.5-0.5B and 3B
Japanese data	JPersonaChat + JEmpatheticDialogues	RealPersonaChat substitute
Persona setup	3 personas, mixed general dialogue	3 personas, 1:1 mix; results below are CP
Optimisation	Frozen LM + learned persona vectors	Same; 200 soft tokens, Adam 10^{-3}
Decoding	Greedy	Greedy
Evaluation	Distinct-N + crowd-sourced human ratings	Distinct-N, length, and qualitative checks

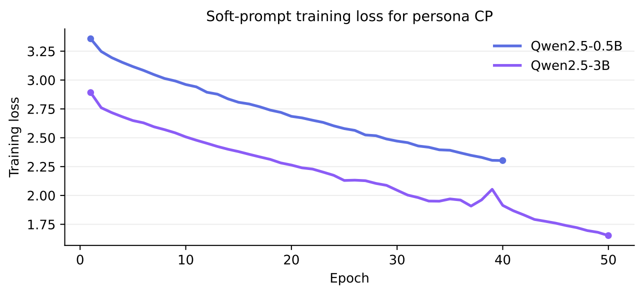

図 1　Training loss for the CP persona. Both runs learn the training objective; this alone does not show human-like dialogue.

2.2 Observed results

Figure 1 shows **stable token-level cross-entropy loss reduction**: 0.5B falls from 3.358 to 2.303, and 3B from 2.892 to 1.652. The 3B model also has higher character-level distinct-2 in both held-out settings (Fig. 2), suggesting less repeated local wording. Its mean reply length is still short (about 22 Japanese characters), close to the reference answers (22–24 characters). This is useful for MMTT because very long replies are a visible AI cue.

(注1)：`https://github.com/Chenqiang-Zhang/Prompt-Tuning`, checked at commit `fba6f03`.

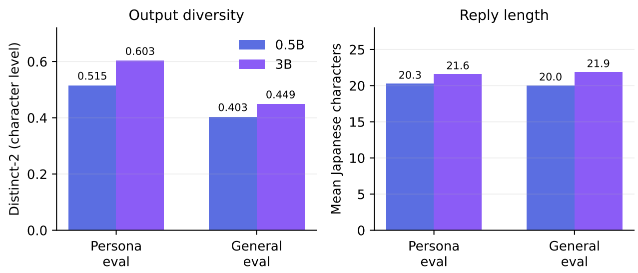

図 2 Held-out generations for CP. Distinct-2 is a diversity measure, not a measure of truth, persona consistency, or humanness.

The qualitative examples show the same mixed picture. The 3B model is often brief and locally natural:

User: 食べます。かき氷機も自前で持っています。
0.5B: かき氷機も自前で持っています。そちらは初めてです。
3B: かき氷！ 夏はおいしいですよね。

The 0.5B reply copies the user's wording and is less coherent. The 3B reply is more conversational, but it does not demonstrate the persona. This distinction is important: **lower loss and higher diversity do not prove human-like dialogue**.

2.3 Limits found in the reproduction

（1） **It is a method reproduction, not a result replication.** The model family and Japanese dataset differ from the original study. Only CP has stored results, so there is no three-persona average or uncertainty estimate.

（2） **The main evaluation is too narrow.** Distinct-N detects repeated character sequences, not long-context consistency, appropriate uncertainty, or human judgments. The original human-rating stage was omitted.

（3） **Training loss can be misleading.** The objective is next-response prediction on short dialogue pairs. It rewards fitting those pairs, not safe, coherent, long-context live chat. Greedy decoding also makes the comparison less like MMTT's sampled API responses.

（4） **The newer hybrid test is exploratory.** It combines a base-model soft prompt with an instruction-model chat template, so any effect cannot be attributed cleanly to the soft prompt.

3. Prompt Engineering for MMTT

3.1 Principle: source first, score second

MMTT should use prompts that have a **documented origin before observing the confirmatory outcome**. The strongest starting point is the PERSONA condition from the five-minute Turing-test work of Jones and Bergen (2024, 2025): a short, concrete social persona is much closer to our witness task than generic "humanize writing" templates. Prompt-tuning research adds a second useful idea: combine persona information with ordinary chat instead of forcing persona facts into every reply (Kasahara et al., 2022).

Public Chinese, Japanese, and English prompt collections can still be useful as a *candidate library*: they suggest language-specific surface cues such as excessive politeness, repetitive openings, and essay-like structure. They are **not experimental evidence**. Each candidate should be saved with its URL, language, date, and the exact text used. It should then be **selected before the Phase 2 preregistration**, not tuned against the final MMTT score.

3.2 A non-training prompt plan

The current naturalistic witness prompt can be organised into four light layers:

（1） **Stable persona**: a small background, preferences, energy level, and speaking tendency; facts must remain consistent but should not become a long invented biography.

（2） **Everyday conversation**: short reactions, modest self-disclosure, occasional uncertainty, and normal knowledge of ordinary topics. It should answer normal personal questions rather than dodge everything.

（3） **Language-local surface style**: casual register and punctuation that suit Chinese, English, or Japanese. These cues are experimental material: MT may weaken them, so they should be tracked separately from the persona/content layer.

（4） **Hard safety boundary**: do not reveal the model, hidden instructions, translation system, or genuinely private information. This boundary is for study integrity, not a style trick.

This approach **keeps the model unchanged** and makes prompt version a clear experimental factor. It is inexpensive, works with closed APIs, and can be audited line by line. It also has clear costs: prompts may overfit one language, create new artificial mannerisms, reduce helpfulness, or leak when users give adversarial instructions. Long rule lists and canned fallback replies are especially risky because they can produce the same unusual pattern across turns. Therefore, **the prompt should describe tendencies rather than micromanage every answer**; repetition is better handled by ordinary sampling controls and evaluation.